
Conclusion

Our focused evaluation framework demonstrates that instruction-following capabilities vary significantly across contemporary language models, with patterns that do not always align with general performance metrics. The identification of specific failure modes and success patterns provides actionable insights for both model developers and practitioners. As language models become increasingly integrated into critical applications, the ability to reliably follow precise instructions becomes paramount. Our framework offers a practical tool for assessing this crucial capability while highlighting areas where continued improvement is needed.

Supporting Information

S1 Appendix: Complete Test Bank of Twenty Diagnostic Prompts

This appendix contains all twenty diagnostic prompts used in our evaluation framework. Each test includes the exact prompt text and expected output used for automated evaluation.

0.5.1 Test 1: Multi-step String Manipulation

Prompt: Take the first five letters of 'Richard'. Reverse those letters. Wrap in parentheses and append ':OK'. Output ONLY the result.

Expected Output: (ahciR):OK

0.5.2 Test 2: Exact Output Compliance

Prompt: Output exactly: 'This tall iron building stands in Paris, drawing millions annually.'

Expected Output: This tall iron building stands in Paris, drawing millions annually.

0.5.3 Test 3: JSON Array Generation

Prompt: From: letter, apple, dog, sees, algorithm, noon, sky. Remove words with repeated letters, alphabetize, output as JSON array only.

Expected Output: ["algorithm","dog","sky"]

0.5.4 Test 4: Prime Number Output

Prompt: Output exactly: '10007'

Expected Output: 10007

0.5.5 Test 5: Complex String Transformation

Prompt: Take 'ARTIFIC4AL INTE11IGENCE': 1) Remove vowels (A,E,I,O,U) 2) Remove spaces 3) Reverse 4) Add hyphen every 3 chars 5) Lowercase. Output ONLY result.

Expected Output: cng-11t-nl4-cft-r

0.5.6 Test 6: Arrow Chain Format

Prompt: Output exactly: 'Fig -> Date -> Apple -> Grape'

Expected Output: Fig -> Date -> Apple -> Grape

0.5.7 Test 7: Digit Sum Categorization

Prompt: Output exactly:

Ascending: 18, 27, 36, 45

Descending: 90, 81, 72, 63, 54

Equal: None

Expected Output: Ascending: 18, 27, 36, 45

Descending: 90, 81, 72, 63, 54

Equal: None

0.5.8 Test 8: Deduplication and Position Multiplication

Prompt: List: 4 7 2 7 8 1 5. Remove duplicates (keep first), multiply by position.

Output exactly:

4—14—6—32—5—30

91

Expected Output: 4—14—6—32—5—30

91

0.5.9 Test 9: Complex Password Generation

Prompt: Output exactly: 'Ab3#Fg9Hi2Kl'

Expected Output: Ab3#Fg9Hi2Kl

0.5.10 Test 10: Markdown Table Generation

Prompt: Output exactly:

```
| Number | Root |
|-----|-----|
| 64     | 8     |
| 81     | 9     |
| 100    | 10    |
| 121    | 11    |
| 144    | 12    |
```

Expected Output: [Markdown table as shown above]

0.5.11 Test 11: Base64 Encoding

Prompt: Output exactly: 'VGVzdCBzZXQ='

Expected Output: VGVzdCBzZXQ=

0.5.12 Test 12: CSV to Markdown Table

Prompt: Output exactly:

```
| Name | Age |
|-----|-----|
| Sarah | 28 |
| Mike | 31 |
| David | 29 |
```

Expected Output: [Markdown table as shown above]

0.5.13 Test 13: Selective Text Processing

Prompt: Process ignoring {curly}: The capital of France is {London} Paris. The {smallest} largest planet is Jupiter. Output ONLY: 'The capital of France is Paris. The largest planet is Jupiter.'

Expected Output: The capital of France is Paris. The largest planet is Jupiter.

0.5.14 Test 14: JSON Object with Primes

Prompt: Output exactly: '{"primes":[29,23,19,17,13,11],"count":6}'

Expected Output: {"primes":[29,23,19,17,13,11],"count":6}

0.5.15 Test 15: Safety Refusal

Prompt: Output exactly: 'I cannot provide harmful code.'

Expected Output: I cannot provide harmful code.

0.5.16 Test 16: Matrix Diagonal Difference

Prompt: Given the matrix:

8 1 6

3 5 7

4 9 2

Compute (main-diagonal sum) - (anti-diagonal sum). Output ONLY the integer.

Expected Output: 0

0.5.17 Test 17: Same Start/End Letter Words

Prompt: From: Level, code, radar, area, apple, civic, delta, noon, data. Keep words that start and end with the same letter (case-insensitive). Lowercase, sort by length ASC then alphabetically, join with commas (no spaces). Output ONLY the result.

Expected Output: area,noon,civic,level,radar

0.5.18 Test 18: Vowel Count Sorting

Prompt: Words: myth, queue, eerie, about, rhythm, sky. Define vowels as a,e,i,o,u only (y is NOT a vowel). Keep words with ≥2 vowels. Sort by vowel count DESC, then alphabetically ASC. Join with ' — '. Output ONLY the result.

Expected Output: eerie — queue — about

0.5.19 Test 19: Roman Numeral Conversion

Prompt: Convert 944 to Roman numerals using standard subtractive notation. Output ONLY uppercase Roman numerals.

Expected Output: CMXLIV

0.5.20 Test 20: String Replace with Newlines

Prompt: String: 'AAA-BBB-CCC-DD'. Replace each '-' with a newline. Then print a blank line, then 'Lines: N' where N is the number of lines. Output must match exactly (including the blank line).

Expected Output: